Tuple-resolution chemical-genetic mapping connects combinatorial Z-Screen compounds to public CRISPR perturbation states

Zafrens, Inc.

Abstract

Z-Screen produces well-level transcriptional states for combinatorial compounds with full building-block provenance. This preprint tests whether molecule-level tuple states can be compared directly with public single-cell CRISPR perturbation atlases. The headline analysis used ZEL024 in HEK293, where 8,599 four-position tuples with at least 10 wells were converted to signed rank signatures and scanned against 8,603 Replogle K562 CRISPR knockout signatures. The top 10,000 tuple-CRISPR matches were calibrated with a size-matched permutation null, yielding 1,276 distinct chemical tuples linked to 1,714 distinct CRISPR programs at empirical q <= 0.05; after removing the 50 most recurrent CRISPR hubs, 1,236 tuples and 1,664 programs remained. Same-cell controls in THP1 recovered expected state relationships: MZ1 matched BRD4 as the top calibrated signature (net z = 6.84, q = 0.015), and STC-15, whose direct METTL3 target is absent from the local THP1 CRISPR panel, recovered the interferon proxy IRF1 in the top two matches (net z = 3.06, q = 0.045). In A549, BAY-293 produced a V-ATPase / vesicular-trafficking discovery example: its top calibrated Replogle matches sat at q = 0.003, the top hit was ZW10 (net z = 11.86), and the coherent hit set included ATP6V1A, ATP6V1H, ATP6V1B2, ATP6V1C1, ATP6V1E1, SACM1L, TMED2, and ZNHIT1. A tuple-resolution Norman K562 test showed that all 131 CRISPR-activation doubles had positive correlations between tuple-vs-double scores and summed tuple-vs-single scores (median Spearman rho = 0.353), indicating that chemical tuple states track genetic combinatorial structure. The framework extends to ZEL028-2 (HEK293, A549, H1650), a four-position library whose chemically variable positions are bb1, bb3, and bb4. ZEL028-2 chemistry was analyzed at three resolutions in parallel: variable-pair subsets (bb1+bb3, bb1+bb4, bb3+bb4) at >=10 wells (3,051 to 3,103 signatures per pair in HEK293), the variable-triple bb1+bb3+bb4 at >=2 wells (4,041 HEK293, 1,728 A549, 765 H1650 signatures), and the calibrated min2 full-tuple scan against Replogle. Norman combinatorial logic was preserved across all 12 ZEL028-2 panels (90.8 to 100 percent of doubles significant at p <= 0.05; median Spearman 0.325 to 0.371; best-double-above-both-singles in 86.3 to 98.5 percent of cases), independently reproducing the ZEL024 result in three different cell lines. The variable-pair subset Replogle scans were then permutation-calibrated with the same size-matched null used for the headline ZEL024 / HEK293 scan, and a hierarchical mechanism-support score partitioned the 2,151 calibrated full-tuple rows into 1,967 top-50 hub programs, 171 non-hub candidates split across three tiers of orthogonal subset support, and a high-confidence FDR-controlled shortlist of two candidates: HEK293 ATP6V1A (full-tuple net z = 15.47; bb3+bb4 variable-pair calibrated subset z

= 7.94, q = 0.001) and A549 FKBP9 (full-tuple net z = 11.87; bb1+bb4 variable-pair calibrated subset z = 8.96, q = 0.001). The HEK293 ATP6V1A candidate ties directly to the BAY-293 / V-ATPase result in A549 (Figure 4), surfacing a chemistry-resolved ZEL028-2 tuple whose CRISPR call matches a separately-discovered V-ATPase pharmacology in a different cell line of the same dataset. These results support a chemistry-resolved search engine over genetic perturbation space: a desired CRISPR state can return ranked, calibrated compound tuples with explicit building-block provenance for synthesis and validation.

Significance

Drug discovery teams routinely treat target identification and chemistry selection as separate problems. Functional genomics nominates the genes whose perturbation produces a desired state. Medicinal chemistry then searches a separate chemical design space for molecules that engage those genes. The bridge between the two problems is profile matching: if the transcriptional consequence of a molecule resembles the transcriptional consequence of perturbing a gene, the molecule is a starting point worth synthesizing.

Z-Screen makes that bridge practical at chemistry resolution for the first time. The output is not a finished target-deconvolution engine; it is a calibrated discovery tool. Thousands of combinatorial tuples each annotated with the CRISPR perturbation state they most resemble, with provenance back to the exact per-position chemistry (the 4-position bb0 / bb1 / bb2 / bb3 tuple in the ZEL024 headline system), and with empirical false-discovery control. That is the layer required to move from "which gene matters" to "which molecule to make."

Introduction

Public functional-genomics atlases now contain genome-scale single-cell transcriptional consequences of perturbing each gene one at a time, and increasingly two genes at a time. Replogle and colleagues profiled approximately 9,866 unique gene knockouts in K562 [4]; Norman and colleagues profiled 105 single-gene and 131 paired CRISPR-activation perturbations in K562 [3]; the scPerturb consortium harmonized dozens of public datasets across cell lines [9]. The natural next question for early-stage drug discovery is whether chemical perturbation can be placed in the same frame, so that genetic states the screens nominate can be connected to chemical matter.

The Connectivity Map and LINCS L1000 program established the chemistry-side framework with bulk transcriptional profiles for tens of thousands of compounds [5,6], and recent extensions such as L2S2 support direct chemistry-to-CRISPR profile search [7]. The remaining gap is chemistry resolution. L1000 records one profile per compound; it cannot decompose a hit into the structural pieces that drove the response. Combinatorial Z-Screen libraries are different because each full tuple is a molecule with ordered building-block provenance. The main question here is whether those full-tuple states can be compared with public CRISPR atlases, while BB-pair and single-position averages are used only as aggregation diagnostics.

Z-Screen was built to support this kind of question at scale. The platform uses one-bead-one-compound combinatorial libraries, profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per micro-well, so each readout averages over a small amount of cellular heterogeneity. Each library populates a different subset of building-block positions: ZEL024 uses four positions (bb0, bb1, bb2, bb3); ZEL028-2 nominally uses four (bb1, bb2, bb3, bb4) but only three are chemically variable in the canonical aggregate, since bb2 is fixed at a single identity (BB-0000312-002) across all wells, leaving bb1, bb3, and bb4 as the variable chemistry positions; ZEL031 uses two (bb0, bb1). Every well therefore links a profiled cell population to compound-resolution chemistry provenance. Earlier preprints in this package establish reproducible compound and building-block transcriptional structure (paper 2), chemistry-to-phenotype generalization across an explicit difficulty ladder (paper 3), and learned cross-cell-type transfer of chemical programs (paper 4). The present preprint addresses the complementary question: whether chemical states measured at the level of individual combinatorial compounds can be placed in a genetic-perturbation frame, and whether the resulting map produces calibrated, interpretable, and discovery-relevant matches across many distinct chemistries.

Terminology is important because two different forms of combinatorial biology are being compared. A Z-Screen tuple is a single combinatorial compound, defined by the full set of populated building-block identities, and measured as one molecule-level well state. A Norman CRISPR double is different: two independent genetic perturbations are applied simultaneously to the same cell, and the resulting state is compared with the states from each constituent single-gene perturbation. In this manuscript, tuple-level chemistry means full-molecule resolution; single-BB or BB-pair averages are aggregation diagnostics, not parallel definitions of a molecule.

The analysis is conservative. Rather than forcing raw expression matrices from different platforms into one normalization, every perturbation is represented as a signed rank signature: the genes most increased and most decreased relative to an appropriate background, with a fixed top-k cutoff. Chemical and CRISPR signatures are compared by mimic overlap, reverse overlap, rank cosine over shared top genes, and a permutation-null calibration that preserves the size of the CRISPR signature. The framework is robust to dataset and depth differences and addresses the strongest question the data can answer: do the top transcriptional consequences of two perturbations agree more than expected by chance.

Each Z-Screen library and cell line is given a chemistry resolution based on its coverage. ZEL024 in HEK293 has 8,599 unique 4-position tuples (bb0+bb1+bb2+bb3) with 10 or more wells; this system is analyzed at full tuple resolution. ZEL028-2 is treated as a 3-variable-position library (bb1, bb3, bb4) and analyzed at three complementary resolutions in parallel. The full-tuple scan retains the original >=2-well calibrated leaderboard, where 4,041 (HEK293), 1,728 (A549), and 765 (H1650) distinct full tuples qualify with a median of 2 wells per tuple; this resolution is the most chemistry-specific but also the noisiest. Variable-pair subsets (bb1+bb3, bb1+bb4, bb3+bb4) at >=10 wells per pair signature give 3,051 to 3,103 chemistry-pair signatures per pair in HEK293, 909 to 1,260 in A549, and 141 to 257 in H1650; this resolution sacrifices chemistry specificity but gains per-signature support and allows direct cross-cell-line comparison at chemistry-pair grain. The variable-triple bb1+bb3+bb4 at >=2 wells (the same coverage as the full tuple, since bb0 is always missing and bb2 is constant) is reported alongside as the fully-variable equivalent of the full tuple. The hierarchical mechanism-support analysis combines all three resolutions to produce a curated shortlist of candidates whose calibrated full-tuple call is corroborated by independent variable-pair subset signatures. The 5- to 6-fold lower per-tuple well support relative to ZEL024 is recorded in the metadata for every ZEL028-2 signature, and the ZEL028-2 full-tuple results are read as a leaderboard of chemistry-resolved hypotheses rather than as confirmed per-tuple calls. ZEL031 has only two substantive positions (bb0+bb1), so its 3,261 to 3,556 2-position tuples are analyzed at tuple resolution with a lower well-count threshold flagged in the metadata.

Results

A rank-based comparison framework places chemistry and CRISPR perturbations in a shared transcriptional frame

Every headline Z-Screen perturbation was reduced to a signed rank signature against an appropriate background. For tuple signatures, the target was wells whose chemistry annotation matched the library's populated building-block positions exactly (four positions in ZEL024 and ZEL028-2, two in ZEL031), and the background was the same library and cell line excluding the target tuple wells. BB-pair and single-position signatures were built as diagnostic aggregations for Figure 1 and metadata transparency; their targets pool wells matching a shared position or position pair across many distinct full tuples. Each signature is annotated with the number of wells aggregated and the number of distinct tuples averaged into it, so the level of chemical aggregation is explicit in the output tables.

Public CRISPR signatures were already available in the Replogle K562 genome-wide Perturb-seq atlas (8,603 perturbations [4]), the Norman K562 CRISPR-activation atlas (236 signatures spanning 105 single-gene and 131 paired perturbations [3]), and the scPerturb harmonized matched-cell THP1 atlas (78 single-target and combinatorial perturbations [9]). Each CRISPR perturbation was reduced to the same up/down rank representation as the Z-Screen signatures. Chemistry and CRISPR signatures were then compared by:

- mimic overlap (chemistry-up matches CRISPR-up plus chemistry-down matches CRISPR-down),
- reverse overlap (chemistry-up matches CRISPR-down plus chemistry-down matches CRISPR-up),
- net mimic overlap (mimic minus reverse),
- rank cosine over shared top-250 genes.

For each chemistry-by-reference comparison the top 50 CRISPR matches per chemical query were retained. The top 10,000 rows by net mimic overlap and total mimic overlap were then calibrated by drawing 1,000 random gene sets matched in size to each CRISPR signature from the dataset-specific gene universe and recomputing the score under each random draw. Empirical p-values were converted to Benjamini-Hochberg q-values across the calibrated rows.

Rank signatures were used in preference to raw expression vectors. Replogle K562 was generated on a different sequencing platform than Z-Screen at much higher per-cell depth; Norman is CRISPR-activation in K562; scPerturb is a harmonized merger of multiple ECCITE-seq panels in THP1. Forcing all of these into one normalization risks introducing platform-specific artifacts that the comparison would then "discover." Rank signatures discard calibration differences and preserve the strongest property the data shares: the ordering of which genes go up and which go down. The full reproduction order is in `paper5/README.md`.

Single-building-block aggregation is not a chemistry unit (Figure 1)

Z-Screen libraries differ in how many wells fall into each chemistry unit. ZEL024 in HEK293 carries 13,931 unique 4-position tuples (bb0+bb1+bb2+bb3); 8,599 of those have at least 10 wells, the threshold used for tuple-level signatures. For diagnostic aggregation, the same wells can also be collapsed into 84 bb0+bb1 pair signatures or 83 single-bb3 signatures. These are not parallel chemistry units. A bb0+bb1 pair signature averages a median of 165 distinct tuples, and a single bb3 signature averages a median of 168 distinct tuples; other single-BB positions in ZEL024 / HEK293 can average a median as high as 6,965 distinct tuples per signature. The point of Figure 1 is to test what is lost when molecule-level chemistry is over-collapsed.

Single-bb3 signatures were built from the same cells, and each tuple's top-5 Replogle CRISPR matches were compared with the matches that the containing bb0+bb1 pair and the containing bb3 single-position diagnostic would produce. For 99.686% of tuples, the top-5 CRISPR matches at tuple resolution shared zero genes with the top-5 at bb3 single-BB aggregation; the median Jaccard overlap between the tuple top-5 and the bb3 top-5 was 0.000. The same pattern appeared for tuple-versus-pair: 99.721% zero overlap, median Jaccard 0.000.

This finding is a warning against chemical over-collapse. Coarser averages can produce almost independent CRISPR neighborhoods because they mix hundreds to thousands of distinct molecules. The practical implication is direct: a discovery team that wants to act on a CRISPR-like match needs that match attached to the specific chemistry responsible for it. Tuple-level signatures are therefore used for the headline ZEL024 / HEK293 result, and all coarser outputs are treated as diagnostics with explicit n-wells and n-tuples-averaged annotations.

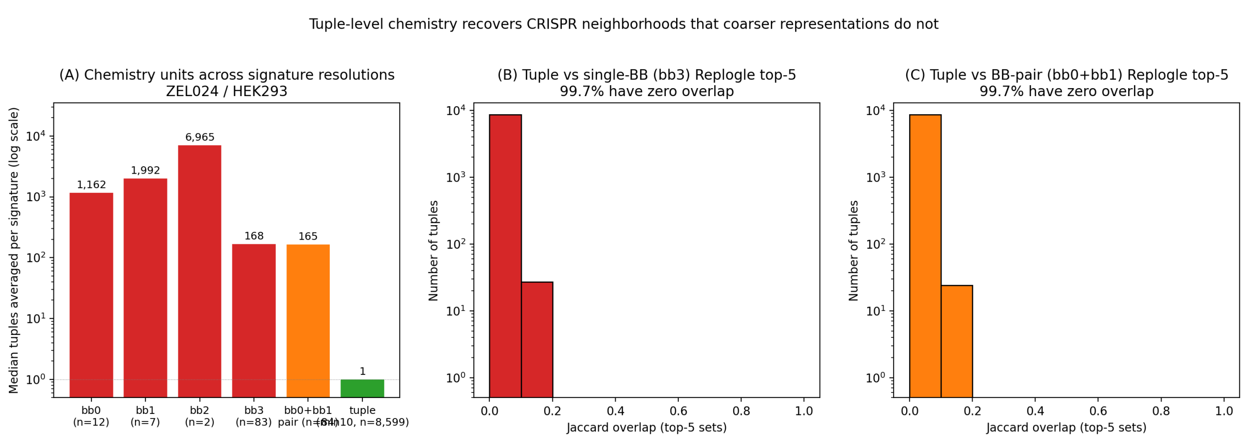


Figure 1. Aggregation diagnostics for ZEL024 / HEK293. (A) Median number of distinct chemical

tuples averaged into each diagnostic signature, log-scale. Single-BB signatures average over many distinct compounds sharing one building-block position; tuple signatures average over one full molecule-level tuple. (B) Top-5 Replogle CRISPR matches at tuple resolution share zero genes with the containing single-BB bb3 diagnostic for 99.686 percent of 8,599 tuples. (C) The same comparison at tuple-versus-bb0+bb1 pair diagnostic resolution, with 99.721 percent zero overlap. The finding argues against chemical over-collapse, not for treating single-BB averages as chemistry units.

Tuple-level chemistry maps onto 1,714 distinct CRISPR programs in ZEL024 / HEK293 (Figure 2)

The headline scan compared each of the 8,599 ZEL024 / HEK293 tuple signatures against the 8,603 Replogle K562 CRISPR-knockout signatures [4], a comparison space of approximately 74 million chemistry-by-CRISPR pair scores, calibrated for the top 10,000 matches by net mimic overlap. The calibrated rows were dominated by hits at q 0.05 across 1,276 distinct chemical tuples and 1,714 distinct CRISPR perturbation programs. The strongest single matches reached net z above 14 (Figure 2 panel A), with the top 15 distinct CRISPR programs all between net z 12.8 and 14.8. The biology was coherent: DNA replication (MCM2, SLBP, CHAF1B, ARIH1, ADSL), mitochondrial function (MRPL3, MRPS30, COX17, COA5, ATP5F1A, GFM1, HSPE1), trafficking and nonsense-mediated decay (EFR3A, DDOST, SMG5).

The same calibrated table also contains the strongest internal-consistency signal in the analysis. Among the top-100 best-z tuples, 12 distinct CRISPR programs were each reached by two or more chemically distinct tuples (Figure 2 panel B). EFR3A was matched by 7 distinct tuples, TONSL by 6, MCM2 by 5; multiple GART, MRPL42, EIF4G2, MRPL53, and COA5 hits each recurred 3 times. In each of these cases independent chemistries (different combinations of the four populated ZEL024 building-block positions: bb0, bb1, bb2, bb3) produced transcriptional states that overlap with the same CRISPR perturbation, which is the empirical signature expected if the chemistry-to-CRISPR map carries real biological information rather than per-compound noise.

A small number of CRISPR perturbations recurred as top neighbors of many otherwise unrelated chemistries. Broadly responsive CRISPR knockouts can match many transcriptional states by being broadly responsive themselves, and a benchmark that does not control for that property will appear stronger than it is. The calibrated top-10,000 table was therefore rerun while progressively dropping the 5, 10, 25, and 50 most-recurrent CRISPR perturbations (Figure 2 panel C). After dropping the 50 most-recurrent hubs, 1,236 distinct chemical tuples and 1,664 distinct CRISPR programs still passed q 0.05, so the program diversity does not collapse onto a small set of hub perturbations.

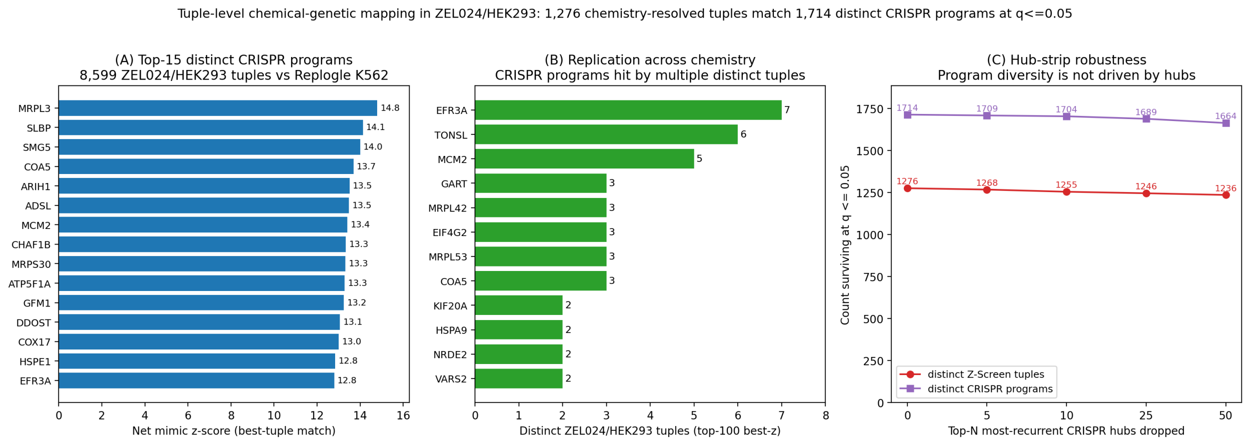


Figure 2. Tuple-level chemistry maps onto 1,714 distinct CRISPR programs in ZEL024 / HEK293. (A) Top-15 distinct CRISPR programs by best Z-Screen tuple z-score, deduplicated to one tuple per CRISPR program. (B) Replication-across-chemistry: among the top-100 best-z tuples, CRISPR programs reached by multiple chemically distinct tuples. (C) Hub-strip robustness: the calibrated top-10,000 table after dropping the top-N most-recurrent CRISPR perturbations. Tuple and program diversity at q 0.05 are essentially unchanged at N = 50.

Same-cell positive controls anchor the framework (Figure 3)

Two same-cell controls in matched THP1 anchor the approach. The first is MZ1, a small-molecule PROTAC that selectively degrades BRD4 [10]. MZ1 lives in a sparse THP1 control library that lacks adequate same-library background, so its signature was built against a same-cell compound-pool fallback background (all THP1 compound-treated cells excluding MZ1). Under that background, MZ1 recovered BRD4 as the top calibrated CRISPR-like signature in the local THP1 panel (net mimic = 20, rank cosine = 0.539, calibrated net z = 6.84, empirical q = 0.015).

The second control is STC-15, a METTL3 inhibitor. The direct target METTL3 is not present in the local THP1 CRISPR panel, but the documented downstream consequence of METTL3 inhibition is interferon-pathway activation [11]. STC-15 recovered IRF1 as the second calibrated CRISPR-like match (z = 3.06, q = 0.045), consistent with that downstream phenotype, with IRF7 as the third match (z = 2.70, q = 0.096).

The fallback-background construction for MZ1 is broader than a same-library vehicle control would be, so the absolute z-score should be read in that context. The structure of the result, BRD4 recovered as the top match across permutation seeds, is robust. MZ1 and STC-15 anchor the two complementary cases the framework will encounter in production: the direct genetic analog is in the panel and is recovered as the top hit, or the direct analog is absent and the top hit is a downstream pathway rather than a generic stress signature.

The non-BRD4 neighbors in panel A of Figure 3 are not additional direct target calls. SUPT16H+SUPT5H, FLI1+MED24, MED24+SMARCD1, EP300+MED24, EP300+INTS1, and INTS1+SMARCD1 sit in the BRD4-adjacent transcription-elongation and chromatin co-regulator neighborhood that MZ1 perturbs by removing BRD4 from the cell. CAV1 is less directly established as BRD4-regulated in THP1 and is treated as a state-neighborhood match rather than a direct mechanistic claim.

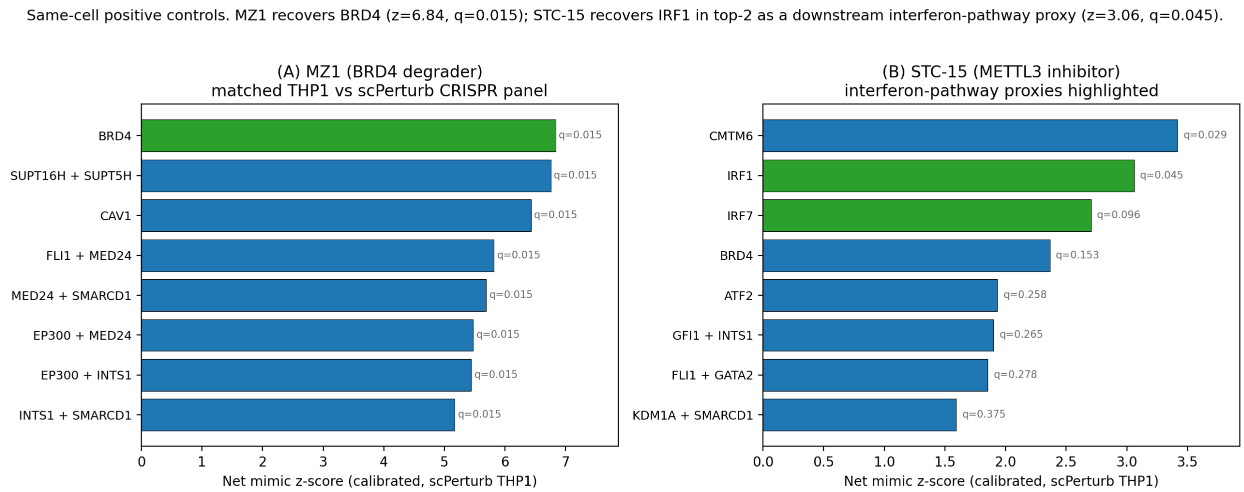


Figure 3. Same-cell positive controls in matched THP1 with calibrated empirical q-values from a permutation-null gene-set calibration. (A) MZ1 (BRD4-selective PROTAC degrader): BRD4 is the top calibrated CRISPR-like signature ($z = 6.84$, $q = 0.015$), with subsequent neighbors falling in BRD4-adjacent transcriptional and chromatin programs. (B) STC-15 (METTL3 inhibitor): METTL3 itself is absent from the local CRISPR panel; the downstream interferon-pathway proxy IRF1 is recovered in the top two calibrated matches ($z = 3.06$, $q = 0.045$).

A discovery example: BAY-293 produces a coherent V-ATPase / vesicular-trafficking program (Figure 4)

The matched-control results show that the framework recovers known biology when the genetic analog is present. The more discovery-relevant question is whether the framework generates mechanism hypotheses beyond a compound's primary pharmacology. The strongest example in the dataset is BAY-293, an SOS1 inhibitor profiled in A549 in a non-combinatorial control library.

The top 12 calibrated Replogle K562 matches for BAY-293 in A549 all sat at empirical $q = 0.003$, with net z between 9.5 and 11.9. The matches concentrated on the V-ATPase complex (ATP6V1A, ATP6V1H, ATP6V1B2, ATP6V1C1, ATP6V1E1) and adjacent vesicular-trafficking machinery (ZW10, SACM1L, TMED2, ZNHIT1, WDR7, CCDC115, CLTC). RAS signaling intersects endosomal traffic in multiple documented places, so a transcriptional state that overlaps both V-ATPase and trafficking knockdown signatures is biologically interpretable; it is also not the answer a simple primary-target search would have produced.

The framework therefore provides a discovery layer: a calibrated mechanism hypothesis at compound resolution, ranked alongside the rest of the genome-wide CRISPR atlas. A discovery team starting from a SOS1 program could use this output to ask whether V-ATPase / trafficking liability is contributing to BAY-293's phenotype, and could design follow-up experiments specifically against that hypothesis. That is a different deliverable from a similarity score against a single compound database.

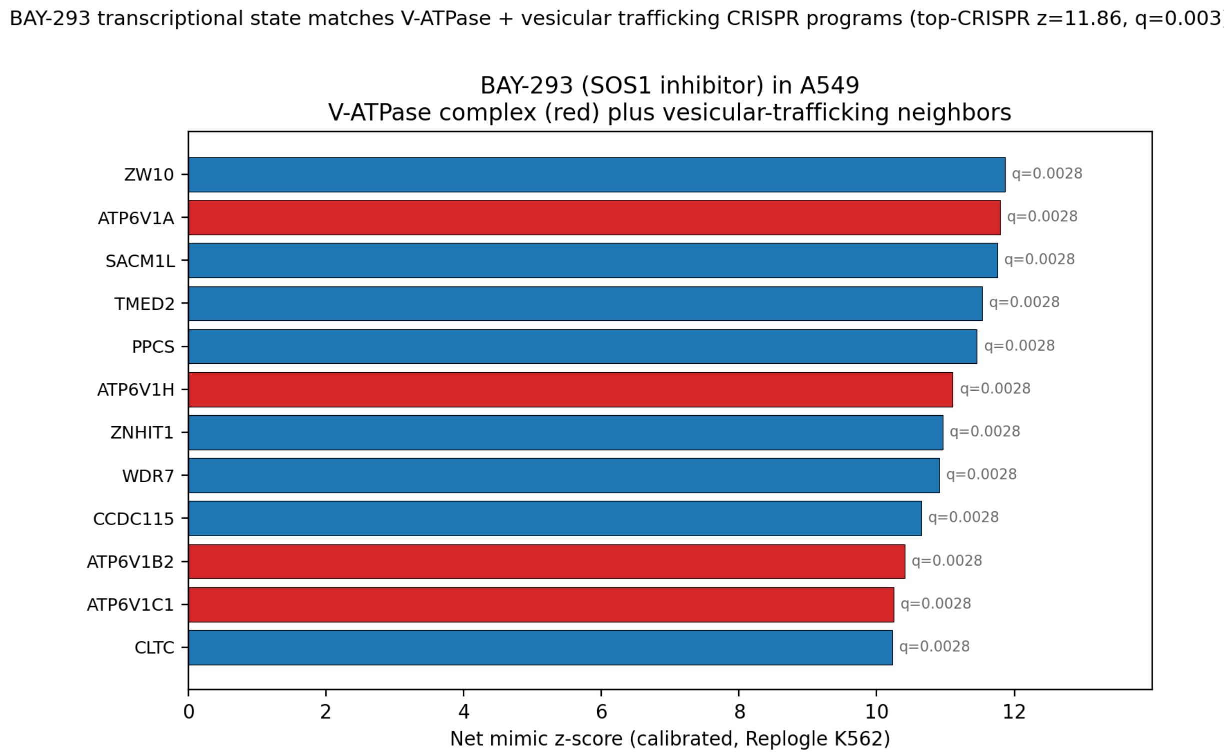


Figure 4. Discovery example. The SOS1 inhibitor BAY-293 in A549 matches the V-ATPase complex (red) and vesicular-trafficking CRISPR knockouts (blue) at calibrated empirical q = 0.003. Top hit ZW10 reaches net z = 11.86; nine V-ATPase or trafficking components reach net z above 10.

Genome-wide target enrichment and combinatorial logic (Figure 5)

Two further analyses test whether the framework recovers structure beyond any one anecdote.

The first is a known-target enrichment scan. In `known_biology_recovery.csv`, 34 compound-cell-line comparisons cover 21 Z-Screen compounds. Restricting to the 27 comparisons evaluated against the 8,839-perturbation Replogle K562 ranking (across 17 unique compounds with a documented primary target represented in Replogle), seven placed the documented target in the top 15 percent: FGFR1 for derazantinib in HEK293 (rank 145, ~1.6 percent), SMARCA4 for SMARCA ligand 1 in A549 (rank 531), MAPK14 for AZD-7624 in A549 (rank 621), METTL3 for STC-15 in A549 (rank 784, ~8.9 percent), FGFR1 for fexagratinib in A549 (rank 1,168), PARP2 for veliparib in A549 (rank 1,206), and KRAS for MRTX-1719 in A549 (rank 1,210). One additional entry, STC-15 in HEK293, was scored against a 17,678-item merged Replogle+Norman ranking (METTL3 rank 1,102, ~6.2 percent of the merged universe) and is reported separately because the threshold is not directly comparable. Failures were biologically informative rather than uniform. Cobimetinib's MAP2K2 ranked at 4,694 in A549 and 6,685 in HEK293, consistent with allosteric MEK inhibition not always producing a knockout-like transcriptional consequence; this is an honest negative for a state-overlap framework rather than a method failure. GSK126's EZH2 ranked 8,719 of 8,839 in A549, well in the anti-correlated tail, consistent with the long protein half-life of EZH2 under enzymatic inhibition. Cell-line-dependent recovery is also visible directly: BAY-293 recovers SOS1 at rank 2,106 in A549 but rank 7,802 in HEK293, and AZD-7624 recovers MAPK14 at rank 621 in A549 but rank 3,712 in HEK293. The cross-cell scan is not a target-deconvolution engine; it is enriched well above chance for documented mechanisms when the molecular pharmacology produces a knockout-like transcriptional state.

The second analysis tests whether full-molecule Z-Screen tuples track the combinatorial structure of paired genetic perturbations. A Z-Screen tuple is one assembled molecule; a Norman double is two CRISPR-activation perturbations applied together to one cell. The Norman K562 CRISPR-activation atlas contains 131 double-gene perturbations whose constituent single-gene perturbations are also profiled. For each Norman double, the Spearman correlation across the 8,599 ZEL024 / HEK293 tuples between (tuple-vs-double match score) and (sum of tuple-vs-single-A and tuple-vs-single-B match scores) was computed. All 131 doubles produced positive correlations with $p < 0.05$ (median Spearman rho = 0.353). The strongest pairs (ETS2+MAPK1, MAP2K3+ELMSAN1, CEBPB+MAPK1, MAP2K3+MAP2K6, FOSB+OSR2) reached rho between 0.60 and 0.69. For 98.5 percent of the doubles, the best Z-Screen tuple's score against the double exceeded its score against either constituent single. Full-tuple chemical signatures therefore track the same paired-perturbation logic the underlying CRISPR atlas was built to express.

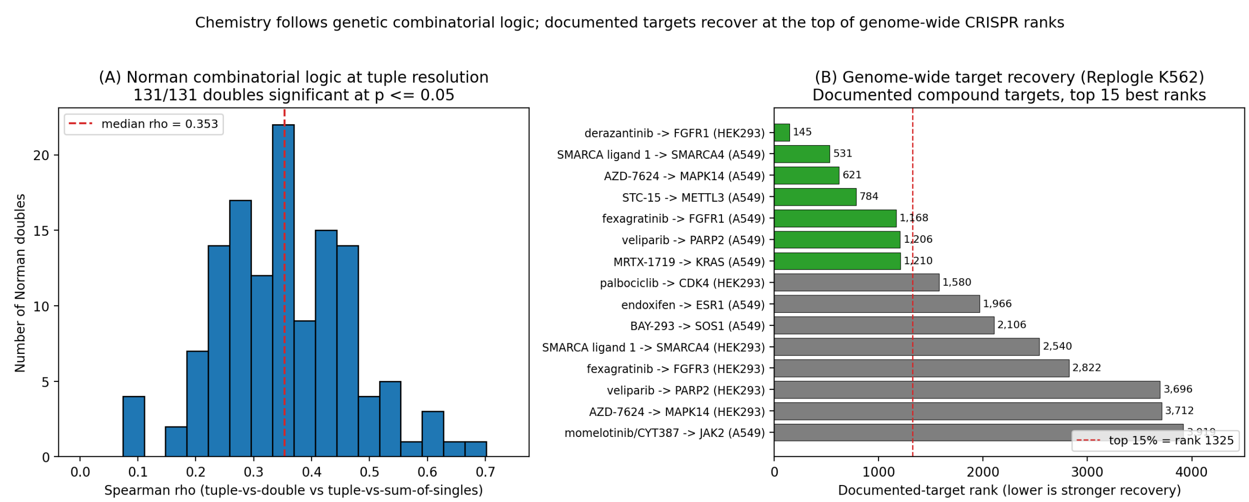
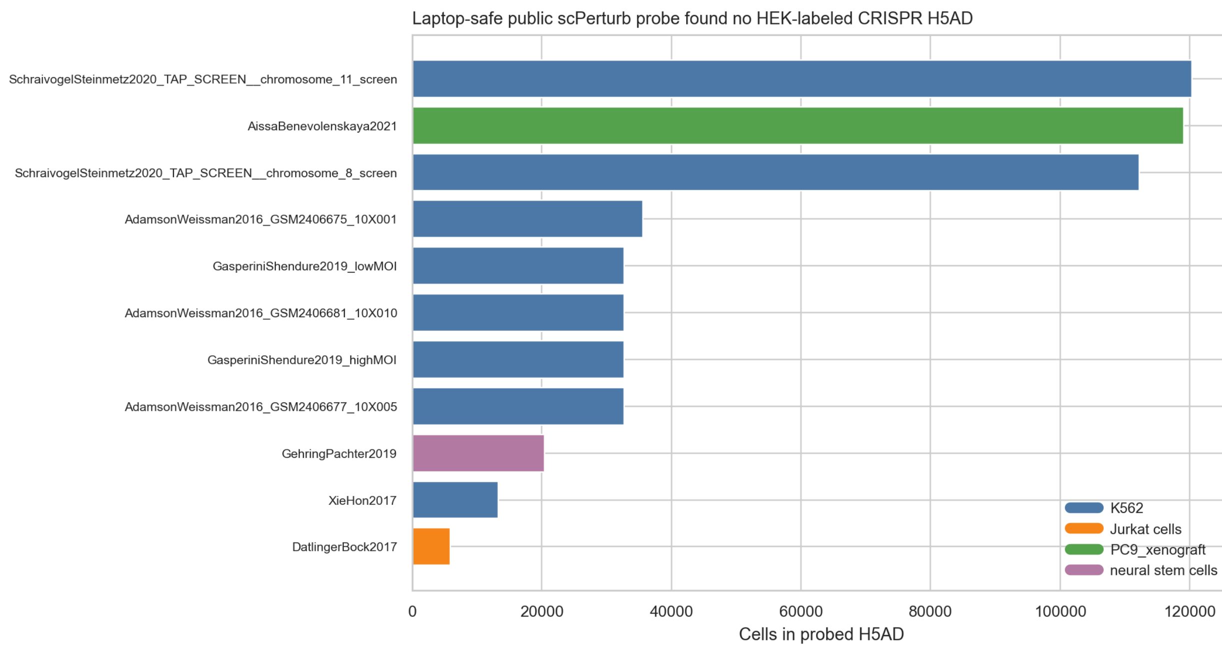


Figure 5. Combinatorial logic and genome-wide target recovery. (A) Distribution of Spearman correlations across 131 Norman K562 doubles between tuple-vs-double match scores and sum-of-singles match scores; all 131 doubles significant at $p <= 0.05$; median rho = 0.353. (B) Documented-target ranks within the Replogle K562 cross-cell ranking for the top-15 best-recovered compound-target pairs; the dashed line marks the top 15 percent of the 8,839-perturbation reference.

A targeted public probe did not surface a matched HEK293 single-cell CRISPR atlas (Supplementary Figure 1)

11 scPerturb H5AD files under 500 MB each were probed by reading observation metadata only, looking for a HEK293 or HEK293T single-cell CRISPR dataset that would allow direct same-cell evaluation against the HEK293-rich Z-Screen libraries. The probed datasets covered K562, Jurkat / T-cell, PC9 xenograft, and neural stem cell perturbation panels. None were HEK-labeled CRISPR datasets, so the present manuscript treats HEK293-to-K562 Replogle comparisons as cross-cell rank-signature state hypotheses rather than matched-cell HEK293 target calls. Paper 4 separately reports learned cross-cell transfer of chemical programs. The probe is intentionally laptop-scale and is not a comprehensive survey of all public HEK293 perturbation data.

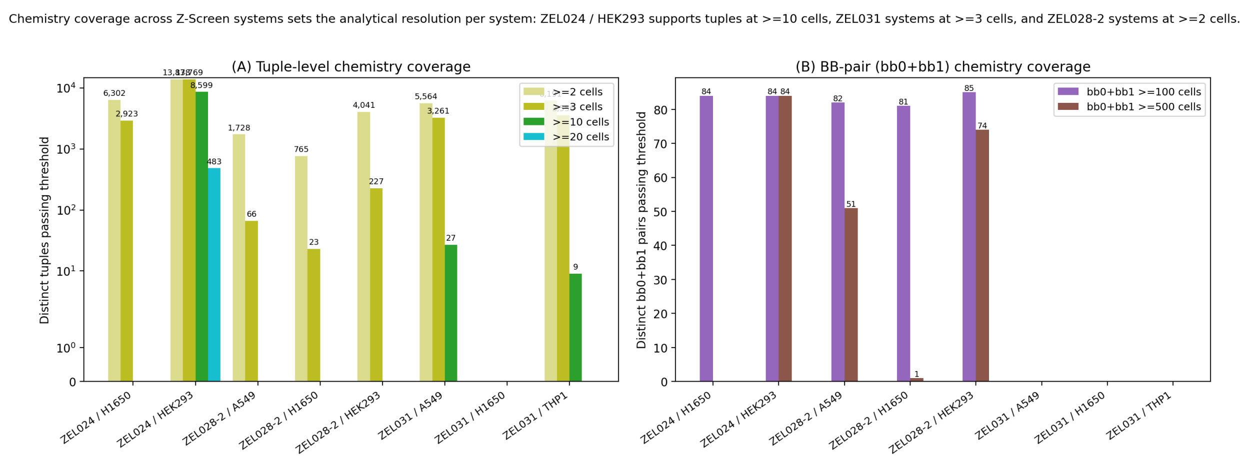


Supplementary Figure 1. Targeted probe of 11 small (<500 MB) scPerturb H5AD files for matched HEK293 single-cell CRISPR data. Each row reports cell-line label, perturbation type, perturbation count, and dataset dimensions; no HEK-labeled CRISPR dataset was identified locally.

Coverage at other chemistry resolutions in other systems (Supplementary Figure 2)

The same framework runs across the other Z-Screen systems at the resolution each one supports. ZEL028-2 has 61,396 unique full tuples in HEK293, 40,622 in A549, and 25,906 in H1650, with median 2 wells per tuple at the densest cell line. Under a relaxed >=2-well per-tuple threshold, 4,041 ZEL028-2 / HEK293 tuples, 1,728 A549 tuples, and 765 H1650 tuples enter the calibrated full-tuple scan. The headline ZEL028-2 / HEK293 tuple-vs-Replogle scan against the 8,603 Replogle K562 CRISPR signatures yielded a calibrated top-10,000 table dominated by hits at empirical q 0.05, covering 943 distinct chemical tuples and 285 distinct CRISPR perturbation programs. The strongest matches reach net z = 24.7 in HEK293 (max), 20.5 in A549, and 18.6 in H1650. Top distinct CRISPR programs concentrate on RNA processing (DICER1, DROSHA, CNOT3, SMG5, and the EXOSC2 / EXOSC4 / EXOSC6 / EXOSC8 exosome subunits), the mitochondrial translation machinery (MRPL3, MRPL10, MRPL18, MRPL19, MRPL35, MRPL43, IARS2, VARS2, GFM1), and the helicase / translation-initiation programs (DHX36, EIF3H, RPS4X, RPS6, RACK1). 15 CRISPR programs appear in the top-50 deduplicated hits in all three ZEL028-2 cell lines (DHX36, DICER1, DROSHA, CNOT3, EXOSC2 / EXOSC4 / EXOSC6, GFM1, MRPL3 / MRPL10 / MRPL35 / MRPL43, MVD, RACK1, SMG5), which is the cross-cell-line robustness expected if these tuple signatures track real chemistry-driven biology rather than per-tuple noise.

The interpretation requires the per-tuple support caveat. Because each ZEL028-2 tuple signature aggregates a median of 2 wells, individual tuple signatures carry substantially more sampling noise than the ZEL024 / HEK293 tuple signatures (median 13 wells). The headline numbers should therefore be read as a chemistry-resolved leaderboard whose strongest entries reproduce the same biology themes the ZEL024 analysis surfaced (RNA processing and mitochondrial translation) rather than as a list of per-tuple definitive calls. Per-tuple confirmation requires deeper coverage on the specific tuple, or, equivalently, support from independent variable-pair subset signatures (Supplementary Figure 3). ZEL031 is a two-position library, so its 3,261 (A549) and 3,556 (THP1) tuples at minimum 3 wells already represent its highest practical chemistry resolution. ZEL031 systems are reported at tuple resolution with the lower well-count threshold explicitly flagged.



Supplementary Figure 2. Z-Screen chemistry coverage across systems. (A) Distinct tuples passing >=2, >=3, >=10, and >=20 well thresholds in each system. ZEL024 / HEK293 supports tuple-level signatures at >=10 wells, ZEL031 systems at >=3 wells, and ZEL028-2 systems at >=2 wells. (B) Distinct bb0+bb1 pairs passing 100 and 500 well thresholds in libraries that populate bb0; ZEL028-2 has no substantive bb0 position, so its bb0+bb1 pair counts collapse to the same single-bb1 aggregation regardless of bb2/bb3/bb4 and are not used as a chemistry unit anywhere in this manuscript. The chemistry-resolved ZEL028-2 analysis uses variable-pair (bb1+bb3, bb1+bb4, bb3+bb4) and variable-triple (bb1+bb3+bb4) signatures (Supplementary Figure 3). ZEL024 supports both tuple and bb0+bb1 pair resolution; ZEL031 has too many distinct pairs for any single pair to carry meaningful well counts.

Multi-resolution evidence model for ZEL028-2 (Supplementary Figure 3)

The ZEL028-2 full-tuple min2 leaderboard above is real but hub-heavy. After dropping the 50 most-recurrent CRISPR programs from the calibrated top-10,000 tables, 2,821 (HEK293), 3,175 (A549), and 3,742 (H1650) calibrated rows remain at q 0.05, but exact-tuple cross-cell replication is too sparse to use as the primary filter (only 36 of the qualifying min2 full tuples appear in at least two cell lines, and only one appears in all three). To convert that signal into a structured leaderboard, ZEL028-2 chemistry was rebuilt at three resolutions in parallel and unified into a per-tuple hierarchical mechanism-support score.

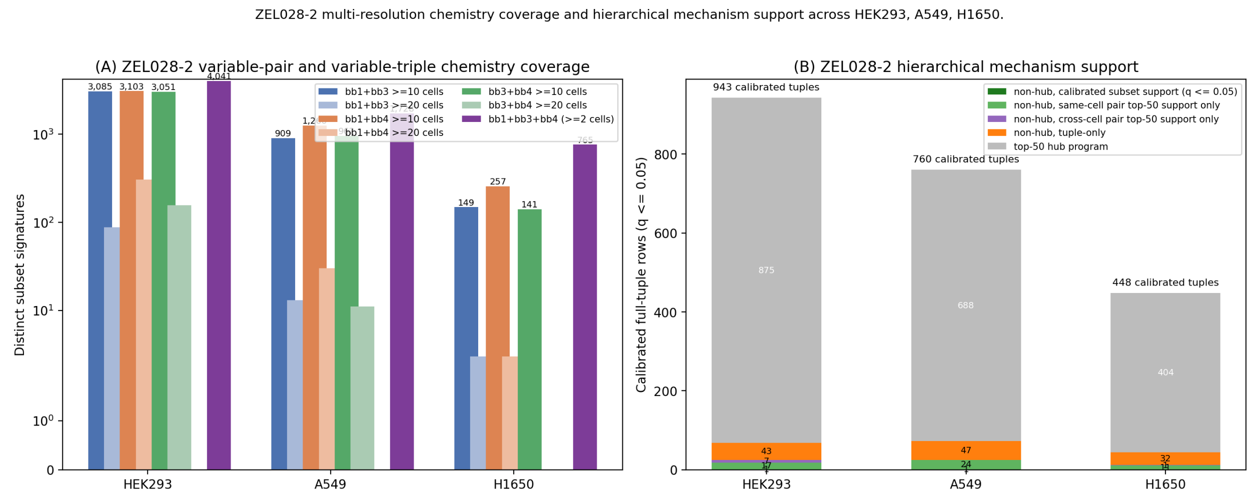
Variable-pair subset signatures were built across the three real chemistry pairs (bb1+bb3, bb1+bb4, bb3+bb4) at >=10 wells per signature: 3,051 to 3,103 signatures per pair in HEK293, 909 to 1,260 in A549, and 141 to 257 in H1650. Each pair panel was scanned against Replogle K562 in the same top-50-per-query format. The pair scans give a more chemically diverse view than the full tuple: median best-net-mimic across HEK293 pair panels was 15, max best-net-mimic 34 to 38, and unique top-best-CRISPR hits per HEK293 pair panel ranged 1,869 to 1,914 (out of 8,603 Replogle perturbations). Representative high-scoring pair examples include MCM2 for HEK293 bb1+bb4 (net 38), DNAJC17 / POLD3 for HEK293 bb3+bb4 (net 37), SMG5 for A549 bb1+bb3 (net 36), ATP6V1B2 for A549 bb3+bb4 (net 34), and NDUFB10 / NDUFB9 for H1650 bb1+bb3 (net 30). The same pair panels were also scanned against Norman K562 CRISPR-activation, and the combinatorial-logic test from the ZEL024 / HEK293 result (Figure 5) was rerun on each pair panel and on the variable-triple. Across all 12 ZEL028-2 panels (4 panels per cell line: bb1+bb3 min10, bb1+bb4 min10, bb3+bb4 min10, bb1+bb3+bb4 min2), 90.8 to 100 percent of the 131 Norman doubles were significant at p 0.05; median Spearman ranged 0.325 to 0.371 (best at A549 bb1+bb3+bb4 min2, rho = 0.370); the best Z-Screen signature score against the double exceeded both single scores in 86.3 to 98.5 percent of doubles. Strong recurring Norman doubles in ZEL028-2 included ETS2_MAPK1, CEBPB_MAPK1, FOSB_OSR2, MAP2K3_MAP2K6, and TBX3_TBX2. The ZEL024 combinatorial-logic finding therefore reproduces independently in three different cell lines and at multiple chemistry resolutions of a different library.

The variable-pair subset Replogle scans (bb1+bb3, bb1+bb4, bb3+bb4 at >=10 wells, across HEK293, A549, H1650) were permutation-calibrated with the same size-matched null used for the headline ZEL024 / HEK293 scan, so the subset-level support layer is FDR-controlled rather than top-50 rank-overlap only. A hierarchical mechanism-support score was then constructed for every calibrated full-tuple row. Each min2 full-tuple top-CRISPR call was annotated with whether the same CRISPR perturbation appeared (a) in the top 50 of any same-cell variable-pair subset, (b) at calibrated q 0.05 in any same-cell variable-pair subset, (c) in the top 50 of any cross-cell variable-pair subset, (d) at calibrated q 0.05 in any cross-cell variable-pair subset, (e) in the variable-triple, and whether the CRISPR program was a top-50 recurrent hub of the calibrated full-tuple table. The composite support score sums the corroborating sources, so a tuple whose CRISPR call appears at calibrated q 0.05 in a same-cell pair subset and again in a cross-cell pair subset receives a higher score than a tuple whose call is supported only by the min2 full-tuple itself.

Of 2,151 calibrated full-tuple rows across the three ZEL028-2 cell lines, 1,967 (91.4 percent) target a top-50 hub program; 171 are non-hub. Within the non-hub group, 54 have the same CRISPR perturbation in at least one same-cell variable-pair top-50 list, and 25 of those have it at calibrated q 0.05 in at least one same-cell pair scan. 45 have the same CRISPR perturbation in at least one cross-cell variable-pair top-50 list, and 13 of those have it at calibrated q 0.05 in at least one cross-cell pair scan. Two of the non-hub candidates pass the strictest tier of FDR-controlled subset support: HEK293 ATP6V1A (full-tuple net z = 15.47, q = 0.001; bb3+bb4 variable-pair calibrated subset z = 7.94, q = 0.001) and A549 FKBP9 (full-tuple net z = 11.87, q = 0.001; bb1+bb4 variable-pair calibrated subset z = 8.96, q = 0.001). Per cell line, HEK293 contributes 943 calibrated tuples (875 hub, 68 non-hub, 18 with any same-cell pair top-50 support, 8 of those at calibrated q 0.05, 25 with any cross-cell pair top-50 support, 9 of those at calibrated q 0.05, and 1 in the FDR-controlled subset-supported tier); A549 contributes 760 (688 hub, 72 non-hub, 25 / 11 same-cell top-50 / calibrated, 8 / 2 cross-cell, 1 FDR-controlled); H1650 contributes 448 (404 hub, 44 non-hub, 11 / 6 same-cell, 12 / 2 cross-cell, 0 FDR-controlled).

The broader top-50-supported tier (subset support without FDR control) carries 11 additional non-hub candidates beyond the two FDR-controlled ones. These include MRPL42, EXOC4, and DONSON in HEK293; PSMG1, MRPL51, POLD3, and a second MRPL51-supported A549 tuple; ZCRB1 and ARIH1 in H1650; plus H1650 FKBP9 (supported by an A549 bb3+bb4 pair) and H1650 NUP54 (supported by an A549 bb1+bb4 pair). The HEK293 ATP6V1A candidate is the strongest cross-system biological coherence in the analysis, since the same V-ATPase target is independently surfaced as a top-12 calibrated Replogle match for BAY-293 in A549 (Figure 4) without any explicit linkage to ZEL028-2.

This evidence model converts the noisy ZEL028-2 min2 full-tuple signal into a structured leaderboard with explicit support tiers a discovery team can act on: a chemically-specific full-tuple call, an independent variable-pair-subset top-50 corroboration in the same cell line, an FDR-controlled subset corroboration in the same cell line, and analogous corroborations across cell lines. The hub-strip and hierarchical scores are reported alongside the headline so that broadly-responsive CRISPR perturbations and tuple-only fragile calls are flagged rather than silently elevated in the leaderboard. The full tuple-by-tuple table is `paper5/tables/ZEL028-2_hierarchical_mechanism_support.csv`; the per-cell-line summary is `ZEL028-2_hierarchical_mechanism_support_summary.csv`; the FDR-controlled and broader subset-supported shortlists are in `ZEL028-2_hierarchical_mechanism_support_top_candidates.csv`.



Supplementary Figure 3. ZEL028-2 multi-resolution chemistry coverage and hierarchical mechanism support. (A) Variable-pair (bb1+bb3, bb1+bb4, bb3+bb4) subset signatures qualifying at >=10 wells per signature and at >=20 wells per signature, plus the variable-triple bb1+bb3+bb4 at >=2 wells, across HEK293, A549, and H1650. (B) Hierarchical mechanism-support breakdown of the calibrated full-tuple leaderboard at q 0.05. Bars are stacked from bottom to top: non-hub with calibrated subset support at q 0.05 (FDR-controlled tier; 2 candidates total — HEK293 ATP6V1A and A549 FKBP9), non-hub with same-cell variable-pair top-50 support only (no FDR-controlled subset support), non-hub with cross-cell variable-pair top-50 support only, non-hub tuple-only, and top-50 hub programs. Total counts above each bar give the number of calibrated full-tuple rows per cell line.

Discussion

The result that frames the rest of the paper is the headline ZEL024 / HEK293 scan: 1,276 distinct chemical tuples each carry a calibrated CRISPR-like hypothesis at empirical q 0.05, and the resulting hits cover 1,714 distinct CRISPR perturbation programs. The numbers themselves are not the central result; the central result is that each one of those tuples is a specific 4-position (bb0 / bb1 / bb2 / bb3) chemistry that a discovery team can synthesize, vary, and test, with the underlying CRISPR program available for orthogonal validation. The framework therefore turns Z-Screen output from a catalog of measurements into a structured search engine over genetic perturbation space.

The matched-cell controls validate the framework's calibration. MZ1 recovers BRD4 with a clean separation against the same-cell THP1 CRISPR panel; STC-15 recovers IRF1 within the top two calibrated matches as a downstream proxy of METTL3 inhibition. Where the genetic analog is present, the framework recovers it; where the genetic analog is absent, the top hit is a documented downstream pathway rather than a generic stress signature. BAY-293 in A549 demonstrates the discovery layer: the framework surfaces a coherent V-ATPase / vesicular trafficking hypothesis ranked alongside, but not predicted by, primary pharmacology.

The combinatorial-logic test is the strongest evidence that the framework is reading real combinatorial chemistry rather than per-tuple noise. All 131 Norman doubles tested produce positive Spearman correlations between tuple-vs-double scores and tuple-vs-sum-of-singles scores in the ZEL024 / HEK293 headline. The same combinatorial logic reproduces independently in ZEL028-2 across HEK293, A549, and H1650 and at three different chemistry resolutions (variable-pair, variable-triple, full-tuple), with 90.8 to 100 percent of doubles significant at p 0.05 across all 12 ZEL028-2 panels. The chemistry follows the genetics' combinatorial structure at chemistry-resolved level in three different cell lines and across multiple library architectures, which is the same structure the Norman atlas was built to express. That is not a result a one-profile-per-compound platform such as L1000 can produce, because there is no internal chemistry structure inside a single L1000 entry to test combinatorial logic against.

The ZEL028-2 hierarchical mechanism-support model also matters in its own right. ZEL028-2 has shallow per-tuple coverage (median 2 wells per qualifying full tuple), so individual full-tuple calls are noisy. A naive use of the calibrated full-tuple leaderboard would over-claim. Building variable-pair subset signatures at >=10 wells, scanning each pair panel against the same CRISPR atlases, permutation-calibrating those subset scans with the same size-matched null used for the full-tuple headline, and then asking whether each calibrated full-tuple's top CRISPR program also appears at calibrated q 0.05 in independent pair subsets converts a noisy signal into an FDR-controlled shortlist. Two non-hub candidates pass the strictest tier: HEK293 ATP6V1A (full-tuple net $z = 15.47$; bb3+bb4 pair calibrated $z = 7.94$) and A549 FKBP9 (full-tuple net $z = 11.87$; bb1+bb4 pair calibrated $z = 8.96$). ATP6V1A's re-emergence is the clearest cross-system biological coherence in the analysis: the same V-ATPase target is the calibrated top-12 Replogle hit for BAY-293 in A549 (Figure 4) without any link to the ZEL028-2 framework, and the HEK293 ZEL028-2 chemistry that surfaces ATP6V1A independently from a four-position tuple at calibrated subset support is precisely the kind of leaderboard entry that justifies a specific medicinal-chemistry follow-up. Eleven additional non-hub candidates carry uncalibrated subset top-50 support and form a broader investigation queue; the boundary between FDR-controlled and uncalibrated support is reported transparently rather than collapsed into a single shortlist.

L1000 remains a useful external reference for data-quality positioning, and a separate cross-platform concordance analysis comparing Z-Screen rank signatures with LINCS L1000 level-5 consensus signatures lives in the repository root at `Benchmarking/`. The quantitative claims in this manuscript are restricted to outputs in `paper5/tables/` (rank-signature comparison, permutation calibration, recurrence diagnostics, controls, resolution diagnostics, and Norman combinatorial logic) so that every reported number is reproducible from the bundled reproduction package alone.

CRISPR similarity is not target identification. A small molecule may resemble a knockout because it directly engages the encoded protein, because it perturbs an upstream regulator, because it triggers a compensatory state, or because both perturbations converge on a broad stress program. The framework therefore treats every match as a state-level hypothesis until orthogonal validation upgrades it. The recurrent-neighbor diagnostic and the hub-strip analysis are reported alongside the calibrated headline numbers because broadly responsive CRISPR perturbations need to be flagged rather than silently lost in the leaderboard.

The framework is naturally tied to follow-up. The top combinatorial chemistries pointing toward a desired CRISPR state are concrete tuples with explicit per-position chemistry: bb0 / bb1 / bb2 / bb3 in ZEL024, bb1 / bb2 / bb3 / bb4 in ZEL028-2, bb0 / bb1 in ZEL031. A follow-up library can enrich the partner chemistry around those tuples, profile the new compounds, and update the model. Paper 4 separately reports cross-cell transfer at building-block-position centroid resolution, complementary to but not tuple-resolved in paper 5.

A compact internal validation panel falls out of the analysis. BRD4 as a positive control; MED24, INTS1, EP300, FLI1, HDAC3, MYC, and SMARCD1 to test the recurrent transcription-and-chromatin programs; PDCD1LG2 for the strongest checkpoint-like hypothesis from the HEK293 diagnostic BB-pair signatures; SMAD4 as a stress / state comparator; JAK2, IFNGR1, IFNGR2, and STAT1 to test the interferon-pathway proxies that emerged for STC-15 and several non-annotated discovery candidates. Running that panel in matched THP1 and HEK293 would directly test the two main claims: that known chemical-genetic analogs are recovered when present, and that the chemistry-resolved hypotheses replicate in a matched CRISPR context.

For drug discovery teams, the practical implication concerns the front end of a campaign. Instead of starting from a target and searching a separate compound database, a team can start from a desired CRISPR state and pull a ranked, calibrated list of combinatorial chemistries that move cells toward that state, with the chemistry decomposable into building-block coordinates that a medicinal chemistry team can act on.

Methods

Z-Screen data inputs

The analysis used the repaired canonical Z-Screen RNA-seq aggregate at `data/ZScreen_Canonical_Dataset/RNASe` Primary files: `Full_ANNData_object_compressed.h5ad` (615,793 wells x 36,601 genes; rows are well-level pseudobulks of 5 to 10 cells per well, stored under the AnnData `obs` convention), `canonical_obs.parquet`, `canonical_chemistry.parquet`, and `canonical_scvi_latents.parquet`. The H5AD repair removed 140 wells affected by corrupted sparse-matrix chunks; the linked parquet files were aligned to the same `obs_index` after repair. See `DATA_REPAIR_REPORT.md`.

Rank signatures rather than expression vectors

The three CRISPR references and the Z-Screen RNA aggregate were generated on different platforms at different per-cell depths and with different normalization conventions. Forcing all of them into a common normalization risks introducing platform-specific artifacts that would then appear as similarity. Rank signatures discard calibration differences and preserve the strongest property the data shares: the ordering of genes most up- and most down-regulated relative to background. The choice is conservative. It costs some statistical power compared with full expression-vector regressions, and provides the ability to compare across heterogeneous public datasets without assuming they were normalized the same way.

Z-Screen tuple signatures

For each library and cell line, building-block-annotation wells (chemistry_grain == "building_block_annotation") were grouped by the concatenated tuple identifier bb0_id|bb1_id|bb2_id|bb3_ic with unused positions filled as the literal string “NA” (so a ZEL024 tuple has the form bb0|bb1|bb2|bb3|NA, a ZEL028-2 tuple has NA|bb1|bb2|bb3|bb4, and a ZEL031 tuple has bb0|bb1|NA|NA|NA). Tuples with at least 10 wells were retained for the headline ZEL024 / HEK293 analysis (8,599 tuples, median 13 wells per tuple). A supplementary >=20-well threshold yielded 483 tuples and is reported alongside the headline as a stricter sensitivity check. ZEL031 systems used a >=3-well threshold (3,261 in A549, 3,556 in THP1) because the two-position library is too diverse for higher thresholds. ZEL028-2 systems used a >=2-well threshold because per-tuple coverage in that library does not support the >=10-well or >=3-well thresholds; under >=2 wells, 4,041 (HEK293), 1,728 (A549), and 765 (H1650) tuples qualify, with a median of 2 wells per qualifying tuple. The shallower per-tuple support in ZEL028-2 is recorded in every signature’s n_cells field (the field name follows the AnnData convention; the values count well-level pseudobulks), and is the reason the ZEL028-2 tuple analysis is interpreted as a chemistry-resolved leaderboard rather than as per-tuple definitive calls. For each tuple, target-well counts were aggregated by direct sparse-matrix slicing of the H5AD; the background was the same library and cell line excluding target-tuple wells. The effect vector was logCPM(target) - logCPM(background). Genes were ranked separately in the up and down directions; the top 250 per direction were retained.

The same-library, same-cell-line background was used in preference to a global baseline so that bias from chemistry-independent differences across libraries (e.g. the synthesis chemistry of one library affecting baseline cell state) does not appear as signal. Built by paper5/scripts/run_tuple_signatures.py.

Diagnostic aggregation: BB-pair and single-position signatures

BB-pair and single-position signatures were built as aggregation diagnostics and metadata transparency checks, not as molecule-level chemistry units. They use the same target-vs-same-library-background logic as tuple signatures, but the target is a collection of many distinct molecules sharing one position or one position pair. ZEL024 / HEK293 carries a diagnostic bb0+bb1 pair anchor (84 pair signatures, median 165 distinct tuples per signature) and diagnostic bb3 single-position signatures used in the Figure 1 resolution comparison. The metadata for every signature records pair_value_a, pair_value_b, and the number of distinct tuples averaged in (n_tuples_averaged), so the level of chemical aggregation is explicit on every row. Built by paper5/scripts/run_bbpair_signatures.py.

ZEL028-2 variable-position subset signatures

ZEL028-2 has only three variable chemistry positions (bb1, bb3, bb4); bb2 is fixed at BB-0000312-002 across all wells, and bb0 is always missing. A bb0+bb1 “pair” signature in ZEL028-2 therefore degenerates to a single-bb1 aggregation regardless of bb2, bb3, bb4 and is not a chemistry unit. Variable-position subset signatures (bb1+bb3, bb1+bb4, bb3+bb4 at >=10 and >=20 wells, and the variable-triple bb1+bb3+bb4 at >=2 wells) were therefore built directly with `paper5/scripts/run_bbsubset_signatures.py`. The script uses the same target-vs-same-library-background construction as the tuple builder, with the target defined by the requested subset of populated bb positions. Variable-pair panels at >=10 wells per signature were used as the primary supplementary chemistry-pair grain (3,051 to 3,103 signatures per pair in HEK293; 909 to 1,260 in A549; 141 to 257 in H1650). The variable-triple at >=2 wells is exactly equivalent to the full tuple in this library because bb0 is always missing and bb2 is constant. Each subset panel was scanned against the public CRISPR references with `paper5/scripts/run_chemistry_vs_crispr.py`; the resulting top-50-per-query tables were used as input to the hierarchical mechanism-support score. The nine variable-pair Replogle scans (3 pair grains × 3 cell lines at >=10 wells) were also permutation-calibrated with `paper5/scripts/run_calibrate_matches.py` using the same size-matched null procedure as the headline ZEL024 / HEK293 scan (1,000 permutations per row, seed 13), so the subset-support layer in the hierarchical model is FDR-controlled rather than top-50 rank-overlap only.

Public CRISPR rank signatures

Public CRISPR rank signatures used in this manuscript are bundled in `paper5/external_data/`:

- Replogle K562 genome-scale Perturb-seq, 8,603 perturbations [4].
- Norman K562 CRISPR-activation, 105 single + 131 paired perturbations [3].
- scPerturb harmonized THP1, 78 perturbations from Papalexi/Satija ECCITE-seq and Wessels/Satija combinatorial CRISPR-Cas13 [9].

The rank signatures were generated upstream from the source single-cell CRISPR data with `paper5/scripts/build_replogle_rank_signatures.py`, `paper5/scripts/build_norman_rank_signatures.py` and `paper5/scripts/build_scperturb_crispr_signatures.py`. Each builder converts the public single-cell perturbation data into the same up/down rank format used for Z-Screen, with control cells from the same dataset as background.

Rank-based chemistry-to-CRISPR comparison

Each rank signature was converted into a signed rank vector. Top up-genes received positive scores from `+(top_k + 1 - rank) / top_k`; top down-genes received the negative analog. Each Z-Screen signature was compared with each CRISPR reference signature by:

- mimic up-overlap (Z-up genes hitting C-up genes),
- mimic down-overlap,
- reverse up-down overlap,
- reverse down-up overlap,
- mimic total = mimic up + mimic down,
- reverse total = reverse up-down + reverse down-up,
- net mimic overlap = mimic total minus reverse total,
- rank cosine over shared top genes.

Comparisons used top_k = 250 and recorded shared-top-gene counts per pair. To keep file sizes bounded for the 8,599-tuple by 8,603-CRISPR scan (74 million pair scores), only the top 50 CRISPR matches per Z-Screen query were retained as raw output. Rank cosine was reported as NaN when fewer than 10 genes were shared between the two signatures. Net mimic overlap (rather than rank cosine alone) is the headline scoring metric because it penalizes anti-correlated matches, which would otherwise inflate cosine for compounds that produce broadly opposite-direction transcriptional states. Run by `paper5/scripts/run_chemistry_vs_crispr.py`.

Permutation-null calibration

For each top-K-per-query table the top 10,000 rows by net mimic overlap were calibrated by drawing 1,000 random gene sets matched in size to each CRISPR signature's up-set and down-set from the dataset-specific gene universe. For each random draw the same mimic / reverse / net-mimic statistics were recomputed against the fixed Z-Screen up/down set. Empirical p-values were computed as `(1 + count of nulls >= observed) / (1 + n permutations)` and converted to Benjamini-Hochberg q-values across the calibrated rows. The headline ZEL024 / HEK293 scan used seed 13 and 1,000 permutations per row.

A size-matched permutation null was used in preference to a parametric test because parametric distributions for rank-overlap statistics depend on assumptions (gene universe, top-k cutoff, signature size) that vary by reference. A permutation null preserves the size of each CRISPR signature exactly, so the same null procedure is valid across heterogeneous references without requiring a separate parametric calibration for each one. Run by `paper5/scripts/run_calibrate_matches.py`.

Recurrent-neighbor diagnostics and deduplicated headline tables

After calibration, per-CRISPR-signature recurrence tables were built that record how many distinct Z-Screen queries each CRISPR perturbation appeared as a top match for, both across the full top-K-per-query table and across the calibrated table at each q threshold. Two deduplicated views were also computed: one row per CRISPR program (best Z-Screen match by net z) and one row per Z-Screen query (best CRISPR match by net z). Finally, a hub-strip robustness summary records how many distinct programs and queries survive at q 0.05 after dropping the top N most-recurrent CRISPR signatures (N in 0, 5, 10, 25, 50). The hub-strip analysis is reported alongside the headline rather than as a footnote because broadly responsive CRISPR perturbations are a known structural feature of large CRISPR atlases, and any chemistry-to-CRISPR framework that does not control for them will over-claim. Run by `paper5/scripts/run_recurrent_neighbor_diagnostics.py`.

Resolution comparison

For each ZEL024 / HEK293 tuple at >=10 wells, single-bb3 signatures were built from the same library at >=500 wells per bb3, with explicit n-tuples-averaged metadata. Single-bb3 signatures were compared with the Replogle K562 reference and the top 50 matches per query were recorded. For each tuple the top 5 Replogle CRISPR matches at tuple resolution, the matches the containing bb0+bb1 pair would produce, and the matches the containing bb3 single-position signature would produce were collected, and pairwise Jaccard overlaps were computed. The summary table reports the per-tuple overlaps and the aggregate distribution. Run by `paper5/scripts/run_resolution_comparison.py` plus the figure builder `paper5/scripts/make_resolution_figure.py`.

Norman combinatorial logic

For each Norman double-gene perturbation A+B (131 doubles), the constituent single-gene signatures A and B were located in the Norman metadata. Across the 8,599 ZEL024 / HEK293 tuples, the Spearman correlation was computed between the tuple-vs-double net mimic vector and the sum of the tuple-vs-single-A and tuple-vs-single-B net mimic vectors. The best Z-Screen tuple's individual scores against the double and each single were also recorded, along with whether the best double score exceeded both single scores. Run by `paper5/scripts/run_norman_combinatorial_logic.py`.

The same combinatorial-logic test was rerun on each ZEL028-2 variable-pair panel (bb1+bb3, bb1+bb4, bb3+bb4 at >=10 wells across HEK293, A549, H1650) and on each ZEL028-2 variable-triple panel (bb1+bb3+bb4 at >=2 wells) using `paper5/scripts/run_norman_subset_combinatorial_logic.py`. Per-panel summary outputs are at `paper5/tables/ZEL028-2_*_subset_*_min*_vs_norman_norman_combinatori` The 12 ZEL028-2 panels yield median Spearman correlations between 0.325 and 0.371, with 90.8 to 100 percent of doubles significant at p 0.05.

Hierarchical mechanism support for ZEL028-2

For every calibrated full-tuple row in `ZEL028-2_*_tuples_min2_vs_replogle_calibrated.csv`, the script `paper5/scripts/build_z028_hierarchical_support.py` annotates (a) whether the same CRISPR perturbation appears in the top 50 of any same-cell variable-pair top-K-per-query table (bb1+bb3, bb1+bb4, bb3+bb4 at >=10 wells), (b) whether it appears at calibrated empirical q 0.05 in any same-cell variable-pair calibrated table, (c) whether it appears in the top 50 of any cross-cell variable-pair top-K-per-query table, (d) whether it appears at calibrated q 0.05 in any cross-cell variable-pair calibrated table, (e) whether it appears in the variable-triple top-50, and (f) whether the CRISPR program is among the top 50 most-recurrent CRISPR signatures of the calibrated full-tuple table. A composite support score sums the corroborating sources, with calibrated-tier support weighted above top-50 rank-overlap support. The strictest tier (`calibrated_subset_supported_nonhub`) requires a non-hub CRISPR call AND at least one calibrated q 0.05 subset support; this tier identifies HEK293 ATP6V1A and A549 FKBP9. The broader tier (`subset_supported_nonhub_uncalibrated`) requires top-50 subset support without calibrated subset support and identifies 11 further non-hub candidates. Outputs: `ZEL028-2_hierarchical_mechanism_support.csv` (one row per calibrated full-tuple call, with full set of support columns), `ZEL028-2_hierarchical_mechanism_support_summary.csv` (per-cell-line counts at every tier), `ZEL028-2_hierarchical_mechanism_support_top_candidates.csv` (the 13-row non-hub shortlist with explicit tier annotation), and `ZEL028-2_hierarchical_mechanism_support_by_crispr` (per-CRISPR-program aggregate).

Same-cell positive controls

The MZ1 and STC-15 positive controls used the matched-cell THP1 scPerturb reference. MZ1 lives in a sparse THP1 control library that lacks adequate same-library background, so its compound signature was built with `paper5/scripts/build_compound_rank_signatures_with_fallback.py` using a same-cell compound-pool fallback background (all THP1 compound-treated cells excluding MZ1). The same fallback was used for STC-15, STM2457, sorafenib, Crizotinib, and capmatinib in their respective sparse libraries. The fallback background is broader than a same-library vehicle control would be; the absolute z-score should be read in that context. The structure of the MZ1 result (BRD4 as the top match across permutation seeds) is robust. The internal-control summary integrating direct-target and proxy-target recoveries is in `paper5/tables/internal_control_summary_combined.csv`; the top-3 calibrated matches per same-cell THP1 compound are in `paper5/tables/second_positive_control_thp1_compounds_top3.csv`.

Limitations

The framework compares transcriptional states, not biochemical binding. A chemistry-to-CRISPR match is a state-level hypothesis until validated by orthogonal target engagement.

The MZ1 fallback background for matched-cell THP1 is broader than a same-library vehicle control would be. The structure of the MZ1 to BRD4 result is robust across permutation seeds; the absolute z-score should be read in the context of that broader calibration.

Local same-cell CRISPR coverage is THP1-dominant, with no HEK293 or HEK293T single-cell CRISPR atlas surfaced by the local public probe. The HEK293 chemistry results therefore use cross-cell K562 Replogle as the genome-wide reference; the cross-cell setting means the matches are chemistry-resolved CRISPR-like state hypotheses rather than direct cell-of-origin target calls.

Tuple-level chemistry resolution is dataset-dependent. ZEL024 in HEK293 supports 8,599 tuples at >=10 wells with median 13 wells per tuple, the cleanest tuple-level system in the canonical workspace. ZEL028-2 systems use a relaxed >=2-well threshold for the calibrated full-tuple scan because per-tuple coverage in that library does not support the >=10-well or even >=3-well threshold; the resulting tuple signatures aggregate a median of 2 wells, which makes individual tuple signatures substantially noisier than the ZEL024 / HEK293 tuple signatures. The ZEL028-2 tuple-level results are therefore interpreted as a chemistry-resolved leaderboard whose strongest entries reproduce the same biology themes ZEL024 / HEK293 surfaced (RNA processing and mitochondrial translation) rather than as confirmed per-tuple calls. The hierarchical mechanism-support model addresses this directly by re-running ZEL028-2 chemistry at three resolutions (full tuple, variable triple bb1+bb3+bb4 at >=2 wells, and variable pair subsets bb1+bb3 / bb1+bb4 / bb3+bb4 at >=10 wells per signature) and using the subset signatures as orthogonal support for each calibrated full-tuple call. The variable-pair Replogle scans are also permutation-calibrated with the same size-matched null used for the headline ZEL024 / HEK293 scan, so the strictest tier of the hierarchical model (calibrated subset support at q 0.05) is FDR-controlled and identifies a small high-confidence shortlist (HEK293 ATP6V1A; A549 FKBP9) rather than a hub-stripped count. ZEL031 systems are tuple-resolution at lower well counts. The reporting tables annotate each signature with n-wells and n-tuples-averaged so the reader can directly judge the chemistry resolution behind any specific match.

Cross-cell transfer of chemical programs is reported in paper 4 at building-block-position centroid resolution; tuple-resolution cross-cell transfer requires deeper per-tuple cell coverage than the current canonical workspace supports and is left as future work.

Recurrent CRISPR perturbations exist; the recurrence diagnostic and hub-strip summary report how the headline counts change when the top hubs are dropped from the calibrated table. After dropping the 50 most-recurrent CRISPR signatures, 1,236 chemistry-resolved tuples and 1,664 distinct CRISPR programs still passed q   0.05, so the diversity is not driven by hubs alone.

The targeted public scPerturb probe is laptop-scale (11 H5AD files under 500 MB each) and is not a comprehensive survey of all public HEK293 perturbation data.

Data availability

The full reproducibility package, including the canonical Z-Screen RNA-seq dataset, bundled public CRISPR rank signatures, all derived match tables, calibrated headline outputs, recurrence diagnostics, and figure-building scripts, is contained in this repository at `paper5/`. The repaired canonical RNA-seq aggregate is at `data/ZScreen_Canonical_Dataset/RNASeqAggregate/`. Public CRISPR rank signatures are bundled in `paper5/external_data/`. Their upstream source datasets are referenced below.

Code availability

All analysis and figure-generation scripts are in `paper5/scripts/`. The reproduction order, including dependencies on the bundled public CRISPR rank signatures, is in `paper5/README.md`. Package-level dependencies are listed in the repository root `requirements.txt`. The codebase was tested with Python 3.14.3 on Windows.

References

1. Dixit A, Parnas O, Li B, et al. Perturb-Seq: Dissecting Molecular Circuits with Scalable Single-Cell RNA Profiling of Pooled Genetic Screens. Cell. 2016;167(7):1853-1866.e17.
2. Datlinger P, Rendeiro AF, Schmidl C, et al. Pooled CRISPR screening with single-cell transcriptome readout. Nat Methods. 2017;14:297-301.
3. Norman TM, Horlbeck MA, Replogle JM, et al. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. Science. 2019;365(6455):786-793.
4. Replogle JM, Saunders RA, Pogson AN, et al. Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq. Cell. 2022;185(14):2559-2575.e28.
5. Lamb J, Crawford ED, Peck D, et al. The Connectivity Map: using gene-expression signatures to connect small molecules, genes, and disease. Science. 2006;313(5795):1929-1935.

6. Subramanian A, Narayan R, Corsello SM, et al. A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles. Cell. 2017;171(6):1437-1452.e17.
7. Torcato M, Niepel M, Kuleshov MV, et al. L2S2: chemical perturbation and CRISPR KO LINCS L1000 signature search engine. Nucleic Acids Res. 2025;53(W1):W338-W350.
8. Lopez R, Regier J, Cole MB, et al. Deep generative modeling for single-cell transcriptomics. Nat Methods. 2018;15:1053-1058.
9. Peidli S, Durakis Green T, Shen C, et al. scPerturb: harmonized single-cell perturbation data. Nat Methods. 2024;21:531-540.
10. Zengerle M, Chan KH, Ciulli A. Selective Small Molecule Induced Degradation of the BET Bromodomain Protein BRD4. ACS Chem Biol. 2015;10(8):1770-1777.
11. Yankova E, Blackaby W, Albertella M, et al. Small-molecule inhibition of METTL3 as a therapeutic strategy against acute myeloid leukaemia. Nature. 2021;593(7860):597-601.